

Nathan Potraz

📍 Redlands, CA, USA ✉ ntpotraz@pm.me ☎ (760)908-9848 📺 in/ntpotraz 🌐 ntpotraz.dev

SUMMARY

Results-driven Software Engineer with 4+ years of experience architecting full-stack solutions. Expertise in Go, C++, and Java backend development, and React/TypeScript frontend. Seeking a Software Engineering role to build innovative solutions.

EXPERIENCE

Hardware Specialist & Small Business Owner

Self-Employed

June 2022 - October 2024, Oceanside, CA

- Engineered and fabricated over **100** custom-tuned competitive gaming controllers, earning an average **5/5 rating across 76 customer reviews**.
- Developed and implemented **intricate modification protocols** to enable advanced features (e.g stick-profile tuning and button remapping) on custom hardware, **demonstrating complex system integration**.
- Optimized device hardware via precision shell modifications to **achieve specific input values out of a total of 352 values** - critical for high-level competitive play.

PROJECT

Devlog

devlog.ntpotraz.dev/ · October 2025 - Present

- Architected a **high-speed backend** (Golang/SQLite/Turso) to enable instantaneous create, edit, and retrieval of log entries. **Sub 1 second** database reads and writes.
- **Implemented CI/CD using GitHub Actions and Docker** to automate building and deploying to Google Cloud Run when a new feature gets merged to main.
- Integrated Clerk for user authorization, supporting SSO (GitHub/Google) and email/password login, **scalable from 1,000s of users to 1,000,000s**.

Japanese-English Translator

github.com/ntpotraz/cs421-translator/ · March 2024 - May 2024

- Engineered a complete C++ natural language processor featuring a **Finite Automata (DFA) Scanner** to tokenize Japanese input and a **Recursive Descent Parser** to enforce strict Subject-Object-Verb (SOV) grammar rules.
- Developed a core translation engine that utilized a **Hash Map data structure** for efficient dictionary lookup and dynamically reordered translated English tokens into the standard Subject-Verb-Object (SVO) structure.
- **Integrated File I/O streams** to process external source files, coupled with specific **error handling routines** to provide detailed feedback on token mismatch or parsing failures in the output.

Hack Compiler

github.com/ntpotraz/VMTranslator/ · github.com/ntpotraz/Assembler/ · March 2020 - April 2020

- Engineered a complete **compilation chain in Java** to translate high-level Virtual Machine code into executable 16-bit binary, utilizing modular **Parser and CodeWriter** logic for efficient instruction mapping and file I/O.
- Developed a **two-pass Assembler** that leveraged a **custom Symbol Table data structure** to successfully resolve user-defined labels and pre-defined symbols before generating the final machine code.
- Built a VM Translator to map stack arithmetic and memory access commands into **optimized Hack Assembly instructions**, validating the entire system against a rigorous test suite to ensure bit-perfect output.

EDUCATION

B.S in Computer Science

California State San Marcos · San Marcos, CA · 3.7 GPA

Relevant Coursework: OOP, Data Structures & Algorithms, Networking, Cloud, Web Programming, Databases, AGILE/SCRUM.

SKILLS

Languages: Go, C++, Java, Python, JavaScript/TypeScript, React, SQL

Tools: Git, GitHub, Linux, Agile/SCRUM, AWS, GCP, Docker, CI/CD
